

Aug. 27, 2026

Numerical Computing II
Assignment VI

Newton Interpolation Polynomial

General instructions

For all computational questions:

1. Implement a general-purpose Python program for Newton interpolation using divided differences. The program should work for any number of data points and should not use built-in interpolation routines.
2. For each problem, construct and display the divided-difference table, obtain the Newton coefficients, and evaluate the interpolating polynomial at the required points.
3. Wherever the exact function is known, perform error analysis by calculating absolute and relative errors and reporting the maximum error where appropriate.
4. Generate suitable plots showing the exact function, interpolating polynomial, data points, and interpolation error. For comparison problems, include the relevant methods/results on appropriate plots.
5. For numerical differentiation questions, obtain the derivative from the Newton polynomial:

$$f'(x) \approx P_n'(x)$$

and implement the differentiation algorithmically rather than using symbolic differentiation routines. Compare with the exact derivative wherever available and perform the corresponding error analysis.

6. Discuss and interpret the numerical results. In particular, comment on the effect of number of data points, polynomial degree, spacing of points, and floating-point errors where relevant.
7. Clearly document the algorithm, code, numerical results, plots, and observations/conclusions for each question.

Q1. Building Newton's Polynomial

Consider $f(x) = x^2 + 2x + 1$

and the data

x	-2	-1	0	1	2
$f(x)$	1	0	1	4	9

- a. Construct the divided-difference table manually.
- b. Obtain the Newton interpolation polynomial $P_4(x)$.
- c. Simplify the polynomial and compare it with the original function.
- d. Write a Python program that:

- constructs the divided-difference table,
- extracts the Newton coefficients,
- evaluates $P_4(x)$,
- evaluates the polynomial at $x = -1.5, -0.5, 0.5, 1.5$

- e. Calculate the absolute error $E(x) = |f(x) - P_4(x)|$.

*The program should work for any number of data points, not just these five.

Q2. Newton Interpolation with Unequally Spaced Data

Consider $f(x) = \sin x$

with the following unequally spaced data:

x	0	0.2	0.55	1.0	1.4
$f(x)$	$\sin(0)$	$\sin(0.2)$	$\sin(0.55)$	$\sin(1.0)$	$\sin(1.4)$

Tasks

- Construct the divided-difference table.
- Construct the Newton interpolation polynomial.
- Evaluate the polynomial at $x = 0.4, 0.8, 1.2$
- Compare each interpolated value with the exact value of $\sin x$.

Calculate $E(x) = |\sin x - P_n(x)|$.

- Plot on the same graph:

- exact function $\sin x$,
- Newton interpolating polynomial,
- original data points.

- Plot the interpolation error $E(x) = |\sin x - P_n(x)|$ over the interval $[0, 1.4]$.

- Answer the following: Why can Newton interpolation be used here even though the points are not equally spaced? & Why would a forward-difference formula based on a constant not be directly applicable to these data?

Q3. Does More Data Always Mean Better Interpolation?

Consider $f(x) = e^x$

using the following 16 unequally spaced points:

$$x_i = [-1.0, -0.92, -0.75, -0.61, -0.48, -0.31, -0.17, 0.02, 0.15, 0.29, 0.43, 0.58, 0.67, 0.81, 0.93, 1.0]$$

Take $y_i = e^{x_i}$

- Write a general Newton divided-difference program and use all 16 points to construct $P_{15}(x)$.
- Evaluate the interpolation at at least 100 points uniformly distributed over $[-1, 1]$.
- Calculate $E(x) = |e^x - P_{15}(x)|$.
- Function comparison plot

Plot: $f(x) = e^x$ and $P_{15}(x)$ on the same graph.

- Error plot: Plot $|e^x - P_{15}(x)|$ against x .
- Report: $E_{\max} = \max \{ |e^x - P_{15}(x)| \}$ (maximum error) the location at which it occurs.

**Your program must construct the polynomial algorithmically from the divided-difference table. Do not manually enter the coefficients of $P_{15}(x)$.*

- Compare the accuracy obtained using:

- first 4 points,
- first 8 points,
- first 12 points,
- all 16 points.

Create a table:

Number of points	Degree	Maximum error
4	3	
8	7	
12	11	
16	15	

- Does increasing the polynomial degree always guarantee a smaller interpolation error? Justify your answer using numerical results.

Q4. Newton vs Lagrange — Are They Really Different?

Consider

$$f(x) = \frac{1}{1+x^2}$$

using the following 17 equally spaced points:

$$x = -4, -3.5, -3, \dots, 3.5, 4$$

Thus, $x_i = -4 + 0.5i \quad i=0, 1, \dots, 16$.

- Implement two independent programs/functions:

1. Lagrange interpolation
2. Newton divided-difference interpolation

- Evaluate both at 200 points over $[-4, 4]$.

- For every evaluation point calculate: $E_N(x) = |f(x) - P_N(x)|$ & $E_L(x) = |f(x) - P_L(x)|$

- Plot

Plot 1: Exact function + Lagrange + Newton

Plot 2: $E_N(x)$ & $E_L(x)$ on the same graph.

- Report: $\max E_N, \max E_L$

- Also compare the interpolated values at: $x = -3.75, -2.25, -0.75, 0.75, 2.25, 3.75$.

- **Answer:** If both methods construct the same interpolation polynomial theoretically, why might their computed answers differ slightly in floating-point arithmetic?

- Also discuss:

- computational cost,
- number of operations,
- ease of adding a new data point,
- numerical error.

Q5. Newton Interpolation and Numerical Error

Investigate the behavior of Newton interpolation as the number of data points increases.

Consider

$$f(x) = \frac{1}{1+25x^2}, \quad -1 \leq x \leq 1.$$

Use equally spaced points:

$$x_i = -1 + \frac{2i}{n}, \quad i = 0, 1, \dots, n.$$

- Construct Newton interpolation polynomials of degrees $n = 4, 8, 12, 16, 20$. (Thus, the largest experiment contains 21 data points.)
- For each n :
 - Construct the Newton interpolation polynomial.
 - Evaluate it at 500 points in $[-1, 1]$.
 - Calculate the maximum absolute error: $E_{\max}(n) = \max_x |f(x) - P_n(x)|$.
- Record the result and Prepare:

n	Number of points	Maximum error
4	5	
8	9	
12	13	
16	17	
20	21	

- Plots

Plot 1: Exact function and interpolation for each selected n .

Plot 2: Maximum error versus n .

Plot 3: Error distribution $|f(x) - P_n(x)|$ versus x , particularly for the largest n .

- Write a short discussion addressing:

1. Does increasing the number of points always improve the interpolation?
2. Where is the error largest?
3. What happens near the endpoints?
4. What happens when the polynomial degree becomes large?
5. How does the choice and distribution of interpolation points affect accuracy?
6. What does this experiment tell you about blindly using a high-degree polynomial?

Q6. Numerical Differentiation Using Newton Interpolation

Consider $f(x) = \sin x$.

Using the data points from Q2, construct the Newton interpolation polynomial. Then obtain the numerical derivative by differentiating the Newton polynomial: $f'(x) \approx P_n'(x)$. Since $f'(x) = \cos x$,

- Compare your numerical derivative with the exact derivative.
- Evaluate it at several points as well, for example, $x = 0.3, 0.7, 1.1$.
- Calculate $E(x) = |\cos x - P_n'(x)|$.
- Required:
 - Compute $P_n(x)$, Differentiate $P_n(x)$, Evaluate $P_n'(x)$. Compare with $\cos x$.
 - Calculate absolute and relative errors.
 - Plot: 1. exact derivative $\cos x$
2. numerical derivative $P_n'(x)$.
 - Plot the derivative error.
- Compare **three methods**: Finite Difference vs Lagrange differentiation vs Newton differentiation

Submission deadline: Aug. 31, 2026